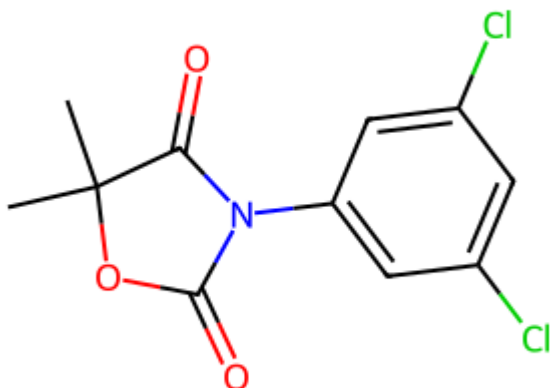


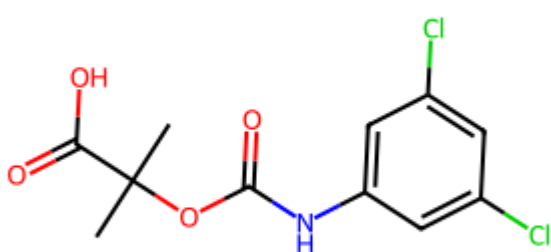


```
{'analysis_time': 0.0,  
'first_solution_time': 0.224972,  
'num_calls_rxn_model': 0,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 8.659535659242902,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 8.659535659242902,  
'retro_star_value': 8.659535659242902}
```



Reaction

```
{'analysis_time': 0.224972,  
'first_solution_time': 0.976809,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'retro_star_min_cost': 8.659535659242902,  
'retro_star_reaction_number': 8.659535659242902,  
'retro_star_rxn_cost': 1.8695340866152856,  
'retro_star_value': 8.659535659242902}
```

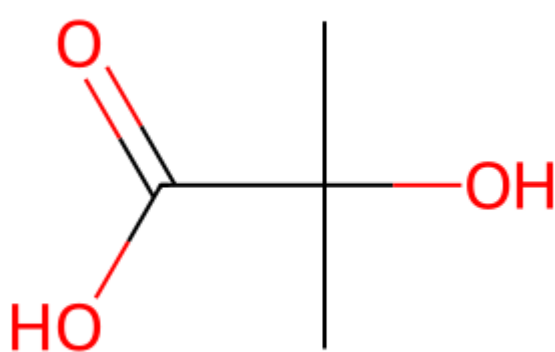


```
{'analysis_time': 0.224972,  
'first_solution_time': 0.976809,  
'num_calls_rxn_model': 1,  
'num_calls_value_function': 0,  
'reaction_number_estimate': 0.0,  
'retro_star_min_cost': 6.790001572627617,  
'retro_star_mol_cost': inf,  
'retro_star_reaction_number': 6.790001572627617,  
'retro_star_value': 8.659535659242902}
```

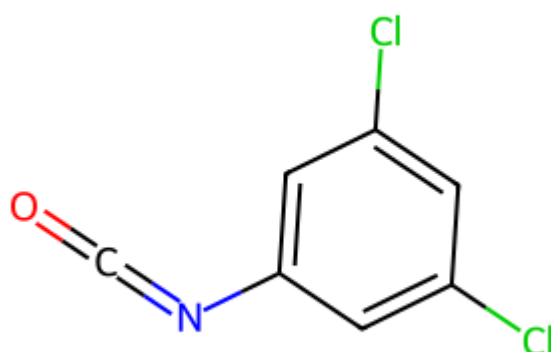


Reaction

```
{'analysis_time': 0.976809,  
'first_solution_time': 0.976809,  
'num_calls_rxn_model': 7,  
'num_calls_value_function': 220,  
'retro_star_min_cost': 6.790001572627617,  
'retro_star_reaction_number': 6.790001572627617,  
'retro_star_rxn_cost': 6.790001572627617,  
'retro_star_value': 8.659535659242902}
```



```
{'analysis_time': 0.976809,  
'first_solution_time': 0.976809,  
'num_calls_rxn_model': 7,  
'num_calls_value_function': 220,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 8.659535659242902}
```



```
{'analysis_time': 0.976809,  
'first_solution_time': 0.976809,  
'num_calls_rxn_model': 7,  
'num_calls_value_function': 220,  
'retro_star_min_cost': 0.0,  
'retro_star_mol_cost': 0.0,  
'retro_star_reaction_number': 0.0,  
'retro_star_value': 8.659535659242902}
```